Introduction to Linear Optimization and Extensions with MATLAB

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Chapter 3

The Simplex Method

Assume LP in standard form

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

General Simplex Method

- Step 0: Generate an initial basic solution $x^{(0)}$. Let $k = 0$, Go to Step 1.
- Step 1: Check optimality of $x^{(k)}$. If $x^{(k)}$ is optimal, then STOP, else go to Step 2.
- Step 2: Check whether the LP is unbounded, if so STOP, else go to Step 3.
- Step 3: Generate another basic solution $x^{(k+1)}$ so that $c^T x^{(k+1)} \leq c^T x^{(k)}$. Let $k = k + 1$, go to Step 1.

Example

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \qquad \le 1 \\ & \qquad x_2 \le 1 \\ & x_1 \ge 0, x_2 \ge 0 \end{aligned}$$

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \qquad + x_3 = 1 \\ & \qquad x_2 \qquad + x_4 = 1 \\ & x_1 \ge 0, x_2 \ge 0, x_3 \ge 0, x_4 \ge 0 \end{aligned}$$

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 + x_3 = 1 \\ & x_2 + x_4 = 1 \\ & x_1 \ge 0, x_2 \ge 0, x_3 \ge 0, x_4 \ge 0 \end{aligned}$$

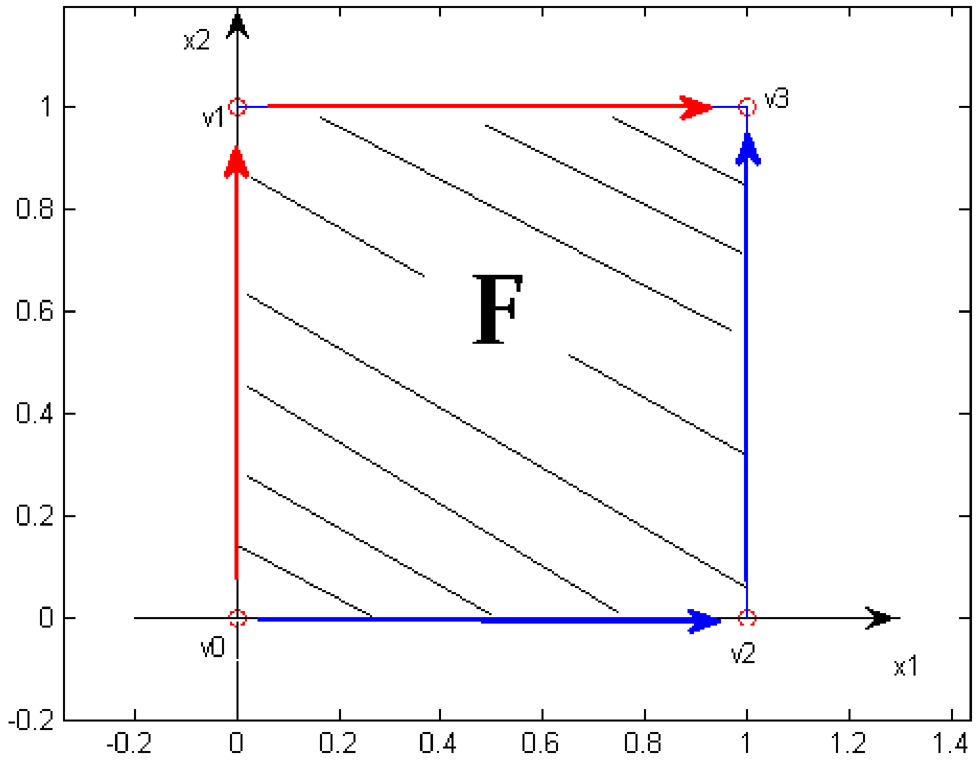
Example

- 4 different extreme points (basic feasible solutions)

$$v^{(0)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad v^{(1)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} x_3 \\ x_2 \\ x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$v^{(2)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad v^{(3)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Example: 2 Dimension



Adjacent Basic Feasible Solution

- $v^{(0)}$ and $v^{(1)}$ are adjacent, $v^{(0)}$ and $v^{(2)}$ are adjacent; $v^{(0)}$ and $v^{(3)}$ are not adjacent.
- $v^{(2)}$ and $v^{(0)}$ are adjacent, $v^{(2)}$ and $v^{(3)}$ are adjacent; $v^{(2)}$ and $v^{(1)}$ are not adjacent.
- Definition: **Two different basic feasible solutions *x* any y are said to be adjacent if they have exactly m-1 basic variables in common.**
- For example, $v^{(0)}$ and $v^{(1)}$ are adjacent since they have m-1=2-1=1 basic variable in common, i.e. x_3

Optimality test for a BFS x*

- Given a basic feasible solution x* how can we determine whether it is optimal without knowing the optimal solution?
- First: What would it mean for x* to be optimal for an LP?

x* optimality

- x* is optimal for an LP if and only if c'x*<=c'x for all x in P (' denotes transpose and P is the feasible set)
- How can we design a test without literally examining all feasible solutions in P?
- Key idea: Since x* is a BFS we can partition x*, A, B according to the basic variables and non-basic variables.
- ***Now rewrite the LP in standard form in terms of the partition.***

Re-write LP according to the partition implied by BFS x*

$$\text{minimize} \quad c^T x$$
$$\text{subject to} \quad Ax = b$$
$$x \geq 0$$

$$x^* = \begin{bmatrix} x_B^* \\ x_N^* \end{bmatrix} \qquad A = [B|N] \qquad c = \begin{bmatrix} c_B \\ c_N \end{bmatrix}$$

$$x_B^* = B^{-1}b$$

For an x in P

x can be re-arranged so that $x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}$

$$c^T x = \begin{bmatrix} c_B \\ c_N \end{bmatrix}^T \begin{bmatrix} x_B \\ x_N \end{bmatrix} = c_B^T x_B + c_N^T x_N$$

The constraints $Ax = b$ can be written as

$$A = \begin{bmatrix} B & N \end{bmatrix} \begin{bmatrix} x_B \\ x_N \end{bmatrix} = Bx_B + Nx_N = b.$$

$x \geq 0$ is equivalent to $x_B \geq 0$ and $x_N \geq 0$.

Partitioned LP

$$\begin{aligned} \text{minimize} \quad & z = c_B{}^T x_B + c_N{}^T x_N \\ \text{subject to} \quad & Bx_B + Nx_N = b \\ & x_B \geq 0, x_N \geq 0 \end{aligned}$$

What does the partitioned LP give us?

- Recall that x^* is optimal iff $c'x^* <= c'x$ for all x in P
- Let $z^*=c'x^*$ and $z = c'x$, then
- x^* is optimal iff $z - z^*$?? 0

Optimality Test

x^* is optimal iff $z - z^* >= 0$

The partitioned LP will help us show under what conditions x^* be optimal i.e.

when $z - z^* >= 0$.

Development of Optimality Test

- From the first constraint:

$$x_B = B^{-1}\left(b - Nx_N\right)$$

minimize $\quad z = c_B{}^T x_B + c_N{}^T x_N$

subject to $\quad Bx_B + Nx_N = b$

$\quad x_B \geq 0, x_N \geq 0$

- Substituting x_B into the objective:

$$\begin{aligned} z &= c_B{}^T B^{-1}\left(b - Nx_N\right) + c_N{}^T x_N \\ &= c_B{}^T B^{-1} b + \left(c_N{}^T - c_B{}^T B^{-1} N\right) x_N \\ &= c_B{}^T x_B{}^* + \left(c_N{}^T - c_B{}^T B^{-1} N\right) x_N \end{aligned}$$

Simplex Method Details – Check Optimal Ratio

$$z - c_B{}^T x_B{}^* = c^T x - c_B{}^T x_B{}^* = \left(c_N{}^T - c_B{}^T B^{-1} N \right) x_N \geq 0$$

- Therefore, the vector $r_N = c_N{}^T - c_B{}^T B^{-1} N$ needs to be non-negative.

- **If $x^* = \begin{bmatrix} x_B{}^* \\ x_N{}^* \end{bmatrix}$ is a basic feasible solution and if the associated reduced costs r_N are non-negative, then x^* is an optimal solution. If there is at least one $r_q < 0$, then x^* is not optimal.**

Example: Optimal Checking

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \qquad + x_3 = 1 \\ & \qquad x_2 \qquad + x_4 = 1 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

For the basic feasible solution $v^{(1)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} x_3 \\ x_2 \end{pmatrix} \\ \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_B = \begin{bmatrix} c_3 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_N = \begin{bmatrix} c_1 \\ c_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$r_1 = c_1 - c_B{}^T B^{-1} N_1 = -1 - (0, -1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -1 < 0$$

$$r_4 = c_4 - c_B{}^T B^{-1} N_4 = 0 - (0, -1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \geq 0$$

$r_1 < 0$, $v^{(1)}$ is not optimal.

Example: Optimal Checking

For the basic feasible solution $v^{(3)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_N = \begin{bmatrix} c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$r_3 = c_3 - c_B{}^T B^{-1} N_3 = 0 - (-1,-1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \geq 0$$

$$r_4 = c_4 - c_B{}^T B^{-1} N_4 = 0 - (-1,-1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \geq 0$$

$r_3 > 0$, $r_4 > 0$, $v^{(3)}$ is optimal.

Simplex Method Details

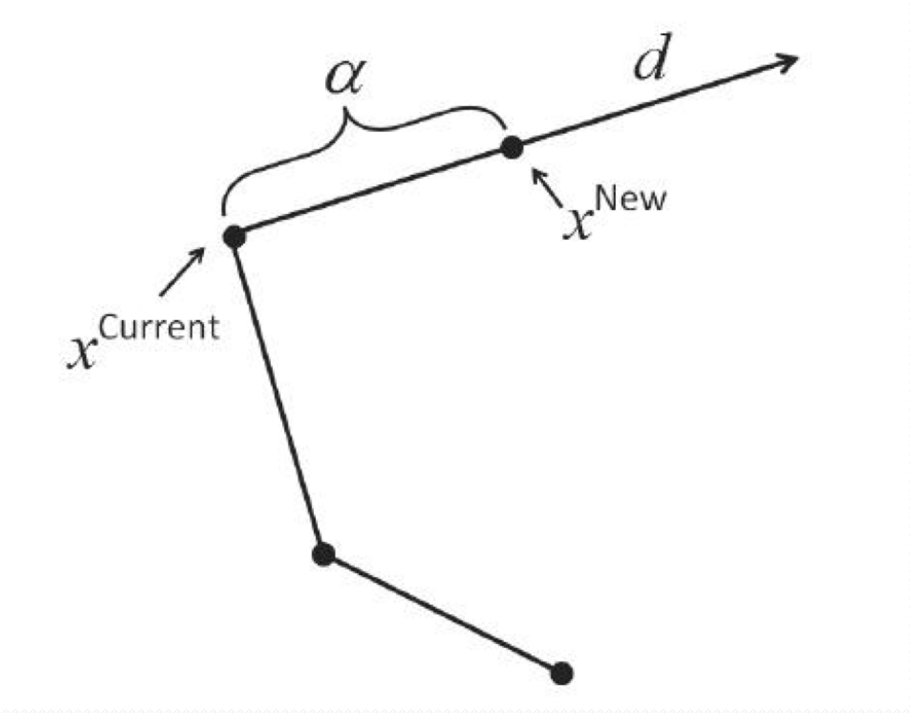
— Moving to an Improved Adjacent Solution

- A move from a current non-optimal basic feasible solution to a new adjacent basic feasible solution can be described as:

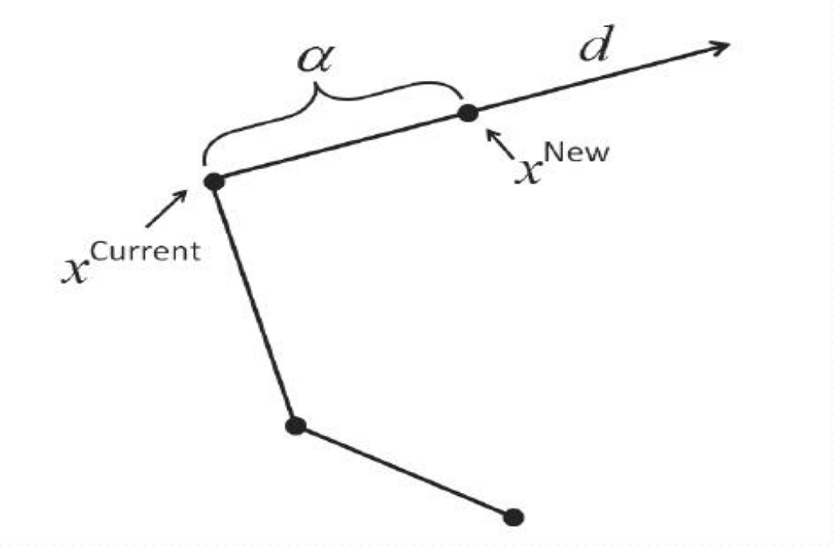
$$x^{new} = x^{current} + \alpha d$$

where $\alpha \geq 0$ is a step length, and d is a direction.

Dynamics of an iteration



Important requirements for selecting direction and step length



- Adjacency requires that the choice of d and α will select one non-basic variable x_q in current non-basic set to become a basic variable in x^{new} , and one basic variable x_l in current basic set to become a non-basic variable in x^{new} .

Simplex Method Details

— Moving to an Improved Adjacent Solution

- From a BFS

$$Bx_B + Nx_N = b \Rightarrow x_B^{current} = B^{-1}b - B^{-1}Nx_N = B^{-1}b$$

to an adjacent BFS .

$$x_B^{new} = B^{-1}b - B^{-1}N_q x_q = x_B^{current} - B^{-1}N_q x_q$$

- Expanding to the full n dimensions of vectors:

$$x^{new} = x^{current} + \alpha d$$

$$\begin{bmatrix} x_B^{new} \\ x_N^{new} \end{bmatrix} = \begin{bmatrix} x_B^{current} \\ x_N^{current} \end{bmatrix} + \alpha d = \begin{bmatrix} B^{-1}d \\ 0 \end{bmatrix} + \alpha d$$

which suggests direction $d = \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix}$, where $e_q = \left.\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}\right\} n-m$ $\rightarrow$ q_{th}

Simplex Method Details

— Moving to an Improved Adjacent Solution

- Theorem 3.1

  Suppose x^* is basis feasible solution with basic matrix B and non-basis matrix N. For any non-basic variable x_q with $r_q < 0$, the direction $d^q = \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix}$ will lead to a decrease in the objective function.

- Proof: $c^T x^{new} = c^T(x^{current} + \alpha d^q) < c^T x^{current}$

  $\Rightarrow c^T d^q < 0$ (α non-negative)

$$c^T d^q = \begin{bmatrix} c_B \\ c_N \end{bmatrix}^T \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix} = -c_B{}^T B^{-1} N_q + c_N{}^T e_q = c_q - c_B{}^T B^{-1} N_q = r_q < 0$$

Simplex Method Details
– Feasibility of a Descent Direction

- Check $x^{new} = x^{current} + \alpha d^q$ satisfy:

  (1) $Ax^{new} = b$ (2) $x^{new} \geq 0$

- To show (1) we consider $Ax^{new} = A(x^{current} + \alpha d^q) = Ax^{current} + \alpha Ad^q = b + \alpha Ad^q = b \Rightarrow Ad^q = 0$.

$$Ad^q = [B \mid N]\begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix} = -BB^{-1}N_q + Ne_q = -N_q + N_q = 0$$

- To show (2) $x^{new} = x^{current} + \alpha d^q \geq 0$

  Case 1: if $d^q \geq 0$, $x^{new} = x^{current} + \alpha d^q \geq 0$ for any $\alpha \geq 0$. but $c^T d^q < 0$ since $r_q < 0$, $c^T(x^{current} + \alpha d^q) = c^T x^{current} + \alpha c^T d^q \to -\infty$ as $\alpha \to \infty$, therefore the LP is unbounded.

Simplex Method Details

– Feasibility of a Descent Direction

Case 2: there is at least one component of d^q that is negative. By the requirement $x^{current} + \alpha d^q \geq 0$ => $\alpha d^q \geq - x^{current}$, thus the largest α is:

$$\alpha = \min_{j \in \bar{B}} \left\{ -\frac{x_j^{current}}{d_j^q} \mid d_j^q < 0 \right\}$$

where $\bar{B}$ is the index set of basic variables in $x^{current}$.

The determination of α is called the minimum ratio test.

Simplex Method Details – Direction and Step Length

- Theorem 3.2

  Consider a LP in standard form and all basic feasible solutions are non-degenerate. Suppose that $x^{current}$ is basis feasible solution that is not optimal, and x_q is a non-basic variable such that the reduced cost $r_q < 0$. Then $x^{new} = x^{current} + \alpha d^q$ is feasible for the LP where $d^q = \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix}$ and α is computed according to the minimum ratio test.

If $d^q \geq 0$, then the LP is unbounded, else x^{new} is a basic feasible solution that is adjacent to $x^{current}$ which has lower objective value i.e. $c^T x^{new} < c^T x^{current}$.

Example: Direction and Step Length

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \qquad + x_3 = 1 \\ & \qquad x_2 \qquad + x_4 = 1 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

$$x^{current} = v^{(1)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} x_3 \\ x_2 \end{pmatrix} \\ \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$r_1 = c_1 - c_B{}^T B^{-1} N_1 = -1 - (0, -1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -1 < 0, \ x^{current} \text{ is not optimal.}$$

$$d^1 = \begin{bmatrix} -B^{-1} N_1 \\ e_1 \end{bmatrix} = \begin{bmatrix} -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} d_3^1 \\ d_2^1 \\ d_1^1 \\ d_4^1 \end{bmatrix}$$

Example: Direction and Step Length

- Since there is at least one component of d^1 is negative, minimum ratio test for α :

$$\alpha = \min_{j \in \bar{B} = \{3,2\}} \left\{ -\frac{x_j^{current}}{d_j^q} \mid d_j^q < 0 \right\} = \left\{ -\frac{x_3^{current}}{d_3^1} \right\} = \left\{ -\frac{1}{-1} \right\} = 1$$

- $x^{new} = x^{current} + \alpha d^q = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_3^{new} \\ x_2^{new} \\ x_1^{new} \\ x_4^{new} \end{bmatrix}$

- $c^T x^{new} = -2 < -1 = c^T x^{current}$

Simplex Method

- Step 0: (Initialization)

  Generate an initial basic solution $x^{(0)} = [x_B \mid x_N]^T$.

  Let B be the basis matrix and N be the non-basis matrix with corresponding partition $c = [c_B \mid c_N]^T$.

  Let $\bar{B}$ and $\bar{N}$ be the index sets of x_B and x_N .

  Let $k = 0$, Go to Step 1.

- Step 1: (Optimality Check)

  Compute the reduced cost $r_q = c_q - c_B^T B^{-1} N_q$ for all $q \in \bar{N}$.

  If $r_q \geq 0$ for all $q \in \bar{N}$, $x^{(k)}$ is optimal, STOP, else select one x_q from non-basis such that $r_q < 0$, Go to Step 2.

Simplex Method

- Step 2: (Descent Direction Generation)

  Construct $d^q = \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix}$.

  If $d^q \geq 0$, then the LP is unbounded, else Go to Step 3.
- Step 3: (Step Length Generation)

  Compute the step length $\alpha = \min_{j \in \bar{B}} \left\{ -\frac{x_j^{current}}{d_j^q} \mid d_j^q < 0 \right\}$ (the minimum ratio test). Let j^* be the index of basic variable then attains the minimum ratio α. Go to Step 4.
- Step 4: (Improved Adjacent Basic Feasible Solution)

  Let $x^{(k+1)} = x^{(k)} + \alpha d^q$. Go to Step 5.

Simplex Method

- Step 5: (Basis Update)

  Let B_{j^*} be the column in B associated with the leaving basic variable x_{j^*}.

  Update the basis matrix B by removing B_{j^*} and adding the column N_q, thus

  $$\bar{B} = \bar{B} - \left\{ j^* \right\} \cup \{q\}$$

  Update the non-basis matrix N by removing N_q and adding B_{j^*}, thus

  $$\bar{N} = \bar{N} - \{q\} \cup \left\{ j^* \right\}$$

  Let k = k+1, Go to Step 1.

Example: Simplex Method

$$\begin{aligned}
\text{minimize} \quad & -x_1 - x_2 \\
\text{subject to} \quad & x_1 \qquad + x_3 = 1 \\
& \qquad x_2 \qquad + x_4 = 1 \\
& x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0
\end{aligned}$$

Step 0: Initialization

$$x^{(0)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ with } B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$c_B = \begin{bmatrix} c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},\ c_N = \begin{bmatrix} -1 \\ -1 \end{bmatrix},\ \bar{B} = \{3,4\},\ \bar{N} = \{1,2\}$$

Example: Iteration 1

- Step 1: $r_1 = c_1 - {c_B}^T B^{-1} N_1 = -1 - (0,0)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -1 < 0$

$$r_2 = c_2 - {c_B}^T B^{-1} N_2 = -1 - (0,0)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 < 0$$

$x^{(o)}$ is not optimal, choose x_1 as the non-basis variable enter the basis. Go to Step 2

- Step 2: Constructed $d^1 = \begin{bmatrix} -B^{-1}N_1 \\ e_1 \end{bmatrix} = \begin{bmatrix} -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \ngeq 0$

- Step 3: Compute step length

$\alpha = \min_{j \in \bar{B} = \{3,4\}} \left\{ -\frac{x_j^{current}}{d_j^1} \mid d_j^1 < 0 \right\} = \left\{ -\frac{x_3^{current}}{d_3^1} \right\} = \left\{ -\frac{1}{-1} \right\} = 1$. Go to Step 4.

Example: Iteration 1

- Step 4:

$$x^{(1)} = x^{(0)} + \alpha d^{(1)} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + (1)\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ x_1 \\ x_2 \end{bmatrix}$$

x_3 leave the basis.

- Step 5: For $x^{(1)}$, $x_B = \begin{bmatrix} x_1 \\ x_4 \end{bmatrix}, x_N = \begin{bmatrix} x_3 \\ x_2 \end{bmatrix}$

Update

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_B = \begin{bmatrix} c_1 \\ c_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, c_N = \begin{bmatrix} c_3 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\bar{B} = \{1, 4\}, \bar{N} = \{3, 2\}$$

Go to Step 1.

Example: Iteration 2

- Step 1: $r_3 = c_3 - c_B{}^T B^{-1} N_3 = 0 - (-1, 0)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \geq 0$

$$r_2 = c_2 - c_B{}^T B^{-1} N_2 = -1 - (-1, 0)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 < 0$$

$x^{(1)}$ is not optimal, choose x_2 as the non-basis variable enter the basis. Go to Step 2

- Step 2: Constructed $d^2 = \begin{bmatrix} -B^{-1}N_2 \\ e_2 \end{bmatrix} = \begin{bmatrix} -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \not\geq 0$

- Step 3: Compute step length

$\alpha = \min_{j \in \bar{B} = \{1,4\}} \left\{ -\frac{x_j^{current}}{d_j^2} \mid d_j^2 < 0 \right\} = \left\{ -\frac{x_4^{current}}{d_4^1} \right\} = \left\{ -\frac{1}{-1} \right\} = 1$. Go to Step 4.

Example: Iteration 2

- Step 4:

$$x^{(2)} = x^{(1)} + \alpha d^{(2)} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + (1)\begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \end{bmatrix}$$

x_4 leave the basis.

- Step 5: For $x^{(2)}$, $x_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, x_N = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$

Update

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, c_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix},, c_N = \begin{bmatrix} c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\bar{B} = \{1,2\}, \bar{N} = \{3,4\}$$

Go to Step 1.

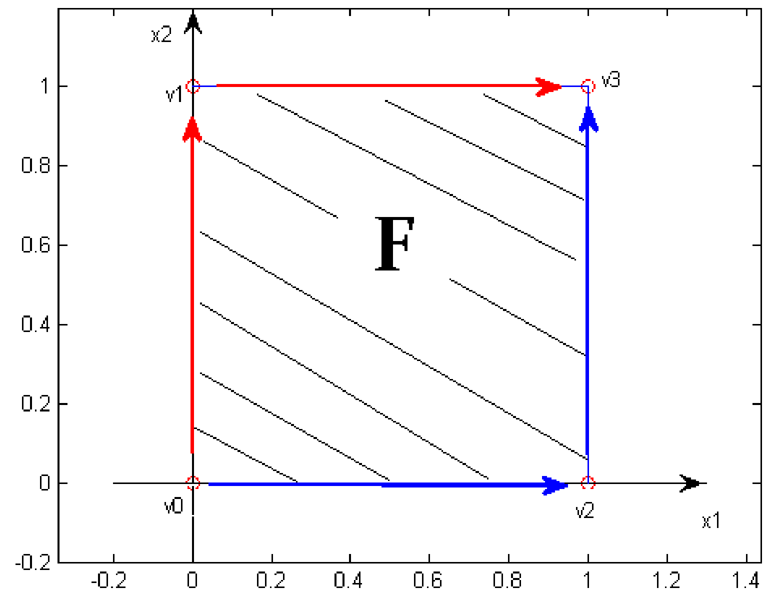
Example: Iteration 3

- Step 1: $r_3 = c_3 - c_B{}^T B^{-1} N_3 = 0 - (-1,-1)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \geq 0$

  $$r_4 = c_4 - c_B{}^T B^{-1} N_4 = 0 - (-1,-1)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \geq 0$$

  $r_N \geq 0$, STOP. $x^{(2)}$ is optimal.

- The path of the iterations is $v^{(0)} \text{->} v^{(2)} \text{->} v^{(3)}$
- If we chose x_2 as enter variable in iteration 1, the path will be $v^{(0)} \text{->} v^{(1)} \text{->} v^{(3)}$



Generating an initial BFS (easy case)

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \phantom{{}+x_2} + x_3 = 1 \\ & \phantom{x_1+{}} x_2 \phantom{{}+x_3} + x_4 = 1 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

We can eyeball an initial BFS.
What variables would you select as initial basic variables? Why?

Generating an initial BFS (easy case)

$$
\begin{aligned}
\text{minimize} \quad & -x_1 - x_2 \\
\text{subject to} \quad & x_1 \quad\quad + x_3 = 1 \\
& \quad\quad x_2 \quad\quad + x_4 = 1 \\
& x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0
\end{aligned}
$$

We can eyeball an initial BFS.
What variables would you select as initial basic variables?
Select the slack variables $\boldsymbol{x_3}$ $\boldsymbol{and}$ $\boldsymbol{x_4}$
Why?

Generating an initial BFS (easy case)

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 \\ \text{subject to} \quad & x_1 \qquad + x_3 = 1 \\ & \qquad x_2 \qquad + x_4 = 1 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

We can eyeball an initial BFS.
What variables would you select as initial basic variables?
Select the slack variables x_3 *and* x_4
Why?
The columns corresponding to x_3 *and* x_4 form an initial basis matrix B that is the identity matrix.

Summary: Easy case

- When constraints are of the form $Ax \leq b$ and $b \geq 0$ and $x \geq 0$
- The standard form version of the constraints is
- $Ax + \mathbf{I}x_s = b$ or $Ax + x_s = b$, $x_s \geq 0$
- In this case, an initial basic feasible solutions is
- $x_B = x_s$ (slack variables) and $x_N = x = 0$ (original variables) with $B = I$ and $N = A$
- Observe that $x_B = B^{-1}b = b \geq 0$

Initial Basis Generation – Two Phase Method

- Artificial variables

  In the case where it is difficult to identity identity sub-matrix in the constraint set $Ax = b$, one can add artificial variables to create an identity sub-matrix.

- E.g. $x_1 + 2x_2 + x_3 + x_4 = 4$

  $-x_1 \quad - x_3 \quad - x_5 = 3$

  The basis is hard to find, we add x_6 and x_7 to the constraints,

$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 \quad + x_6 \quad &= 4 \\ -x_1 \quad - x_3 \quad - x_5 \quad + x_7 &= 3 \end{aligned}$$

$$A = \left[\begin{array}{ccccc|cc} 1 & 2 & 1 & 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 & -1 & 0 & 1 \end{array}\right], b = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Initial Basis Generation – Two Phase Method

- Consider a standard form of a Linear Programming (SFLP)

$$\begin{aligned} &\text{minimize} \quad c^T x \\ &\text{subject to} \quad Ax = b \\ &\qquad\qquad\quad x \geq 0 \end{aligned}$$

And consider the addition of artificial variables (ALP)

$$\begin{aligned} &\text{minimize} \quad c^T x \\ &\text{subject to} \quad Ax + x_a = b \\ &\qquad\qquad\quad x \geq 0 \end{aligned}$$

- The relationship between SFLP and ALP is as follows:

  (1) If SFLP has a feasible solution, then ALP has a feasible solution with $x_a = 0$.

  (2) If ALP has a feasible solution with $x_a = 0$, then SFLP has a feasible solution.

Initial Basis Generation – Two Phase Method

- Phase I: minimize $1^T x_a$

  subject to $Ax + x_a = b$

  $x \geq 0, x_a \geq 0$

  The Phase I problem can be initialized with the basic solution $x = 0$, $x_a = b$.

  Let $\bar{x} = \begin{bmatrix} x^* \\ x_a^* \end{bmatrix}$ be the optimal solution to Phase I.

- Case 1: If $x_a^* \neq 0$, then the original LP is infeasible.
- Case 2: otherwise $x_a^* = 0$ with 2 sub-cases:

Initial Basis Generation – Two Phase Method

- Subcase 1: All x_a^* in the non-basis, then discard the artificial variables in ALP and use the x^* as a starting solution for Phase II problem:

$$\begin{aligned} \text{minimize} \quad & c_B^T x_B + c_N^T x_N \\ \text{subject to} \quad & Bx_B + Nx_N = b \\ & x_B \ge 0, x_N \ge 0 \end{aligned}$$

- Subcase 2: Some of the artificial variables are basic variables. Then exchange such an artificial basic variable with a current non-basis and non-artificial variables x_q. Let x_a^{*i} be such artificial variable in the jth position among x_B. There are 2 sub-subcases:

Initial Basis Generation – Two Phase Method

- Sub-subcase 1: If there is an x_q such that $e_j^T B^{-1} N_q \neq 0$, then x_q can replace x_a^{*i} in the Phase I optimal basis.
- Sub-subcase 2: If $e_j^T B^{-1} N_q = 0$ for all non-basic x_q, then the jth row of the constraint matrix A is redundant and so can be removed, and Phase I restarted.

Example: Two Phase Method

$$
\begin{aligned}
\text{minimize} \quad & x_1 + 2x_2 \\
\text{subject to} \quad & x_1 + x_2 \geq 2 \\
& -x_1 + 2x_2 \geq 3 \\
& 2x_1 + x_2 \leq 4 \\
& x_1 \geq 0, x_2 \geq 0
\end{aligned}
$$

- Phase I:

$$
\begin{aligned}
\text{minimize} \quad & x_6 + x_7 + x_8 \\
\text{subject to} \quad & x_1 + x_2 - x_3 + x_6 = 2 \\
& -x_1 + 2x_2 - x_4 + x_7 = 3 \\
& 2x_1 + x_2 + x_5 + x_8 = 4 \\
& x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0
\end{aligned}
$$

Example: Two Phase Method

$$x_B = \begin{bmatrix} x_6 \\ x_7 \\ x_8 \end{bmatrix} = b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, c_B = \begin{bmatrix} c_6 \\ c_7 \\ c_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_N = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, c_N = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, N = \begin{bmatrix} 1 & 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 & 5 \end{bmatrix}$$

$$\bar{B} = \{6,7,8\}, \bar{N} = \{1,2,3,4,5\},$$

- Solving Phase I by simplex method gives optimal solution

$$x_B^* = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, x_N^* = \begin{bmatrix} x_4 & x_5 & x_6 & x_7 & x_8 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

Example: Two Phase Method

- $x_a^* = [x_6, x_7, x_8]^T = [0\ 0\ 0]^T$, and are all in non-basis, then reformulate Phase II problem:

$$\begin{aligned}
\text{minimize} \quad & x_1 + 2x_2 \\
\text{subject to} \quad & x_1 + x_2 - x_3 = 2 \\
& -x_1 + 2x_2 - x_4 = 3 \\
& 2x_1 + x_2 + x_5 = 4 \\
& x_1, x_2, x_3, x_4, x_5 \geq 0
\end{aligned}$$

- And use the initial basic feasible solution x* from Phase I, we finally get the optimal solution:

$x_1 = 1/3$, $x_2 = 5/3$, $x_3 = 1$, $x_4 = 0$, $x_5 = 5/3$,

Initial Basis Generation – Big M Method

- The Big M problem is

$$\begin{aligned} \text{minimize} \quad & c^T x + M 1^T x_a \\ \text{subject to} \quad & Ax + x_a = b \\ & x \geq 0, x_a \geq 0 \end{aligned}$$

where M > 0 is a large parameter.

- The Big M problem can be solved by the Simplex Method with the initial basic feasible solution $x_a = b$ as basic variables and $x = 0$ as the non-basic variables.
- One of two cases will be result using theBig M method:

Initial Basis Generation – Big M Method

- Case 1: Solving the Big M model results in a finite optimal solution $x = \begin{bmatrix} x^* \\ x_a^* \end{bmatrix}$

  Subcase 1: If $x_a^* = 0$, then x* is the optimal for the original LP.

  Subcase 2: If $x_a^* \neq 0$, then the original LP is infeasible.

- Case 2: The Big M problem is unbounded below.

  Subcase 1: If $x_a^* = 0$, then the original LP is also unbounded below.

  Subcase 2: If at least one artificial variable is non-zero, then the original LP is infeasible.

Example: Big M Method

$$
\begin{aligned}
\text{minimize} \quad & x_1 + 2x_2 + Mx_6 + Mx_7 + Mx_8 \\
\text{subject to} \quad & x_1 + x_2 - x_3 + x_6 = 2 \\
& -x_1 + 2x_2 - x_4 + x_7 = 3 \\
& 2x_1 + x_2 + x_5 + x_8 = 4 \\
& x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0
\end{aligned}
$$

- Solving by Simplex method, the optimal solution is:

  $x_1 = 1/3$, $x_2 = 5/3$, $x_3 = 0$, $x_4 = 0$, $x_5 = 5/3$,

  $x_6 = 0$, $x_7 = 0$, $x_7 = 0$.
- Since $x_a{}^* = [x_6, x_7, x_8]^T = [0\ 0\ 0]^T$, the solution is optimal to original LP.

Degeneracy

- It is possible for basic feasible solutions to have some basic variables with zeros values i.e. the basic feasible solution is degenerate.
- Suppose x_d be a degenerate basic feasible solution, and $x_{new} = x_d + \alpha d^q$ is feasible, i.e.

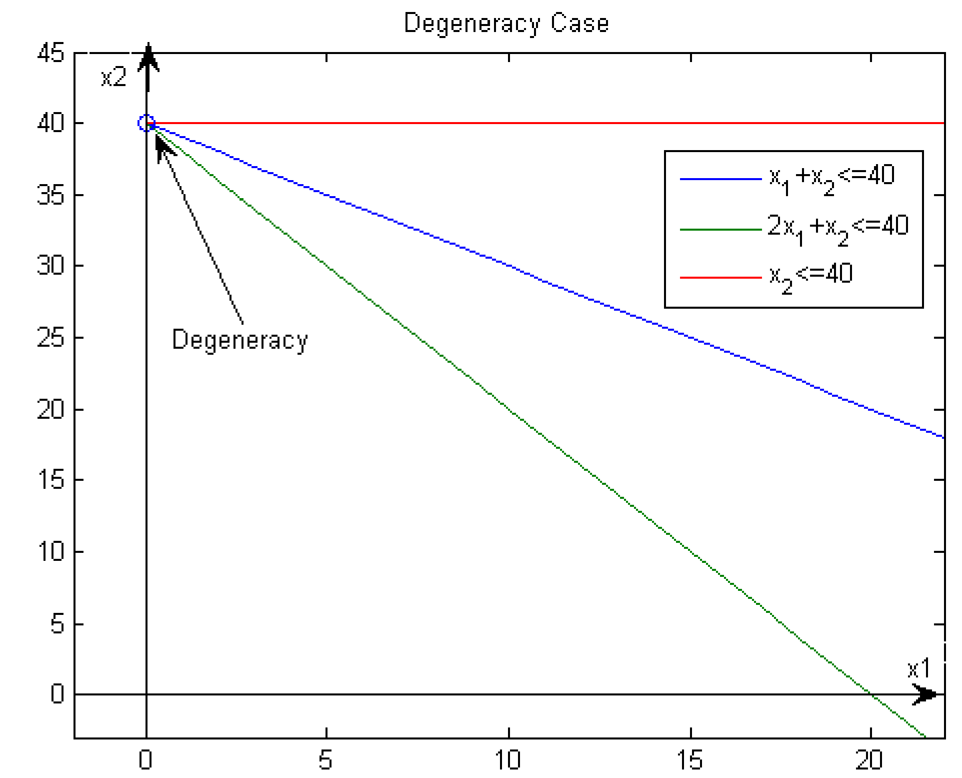
$$Ax_{new} = \mathrm{b} \qquad x_{new} = x_d + \alpha d^q \geq 0$$

since some of the basic variables in x_d are zero, then the minimum ratio test may set α to 0. In such case, the value of x_{new} and x_d *is the same, and there is no improvement in objective value.* However, x_{new} is distinct from and adjacent to x_d.

Example: Degeneracy

- Consider the constraints

$$x_1 + x_2 \le 40$$
$$2x_1 + x_2 \le 40$$
$$x_2 \le 40$$
$$x_1 \ge 0 \quad x_2 \ge 0$$



Cycling

- It is possible that the Simplex Method return a basic feasible solution that was visited in the previous iterations, i.e. cycling occurs.
- Beale's (1955) example

$$
\begin{aligned}
\text{minimize} \quad & -\tfrac{1}{4}x_4 + 20x_5 - \tfrac{1}{2}x_6 + 6x_7 \\
\text{subject to} \quad & x_1 + \tfrac{1}{4}x_4 - 8x_5 - x_6 + 9x_7 = 0 \\
& x_2 + \tfrac{1}{2}x_4 - 12x_5 - \tfrac{1}{2}x_6 + 3x_7 = 0 \\
& x_3 + x_6 = 1 \\
& x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0
\end{aligned}
$$

Cycling

- Using initial B associated with x_1, x_2 and x_3, the following iterations are obtained:

iteration	Enter variable	Leaving variable	Basic variables	Obj val
0			$x_1=0$, $x_2=0$, $x_3=1$,	0
1	x_4	x_1	$x_4=0$, $x_2=0$, $x_3=1$,	0
2	x_5	x_2	$x_4=0$, $x_5=0$, $x_3=1$,	0
3	x_6	x_4	$x_6=0$, $x_5=0$, $x_3=1$,	0
4	x_7	x_5	$x_6=0$, $x_7=0$, $x_3=1$,	0
5	x_1	x_6	$x_1=0$, $x_7=0$, $x_3=1$,	0
6	x_2	x_7	$x_1=0$, $x_2=0$, $x_3=1$,	0

Anti-Cycling Rules – Bland's Rule

- Bland's Rule:

  (1)For non-basic variables with negative reduced costs, select the variables with smallest index to enter the basis.

  (2)If there is a tie in the minimum ratio test select the variables with smallest index to leave the basis.

- Bland (1977) proof that if the Simplex Method uses Bland's Rule, then the Simplex Method will not cycle.

Anti-Cycling Rules – Bland’s Rule

iteration	Enter variable	Leaving variable	Basic variables	Obj val
0			x_1=0, x_2=0, x_3=1,	0
1	x_4	x_1	x_4=0, x_2=0, x_3=1,	0
2	x_5	x_2	x_4=0, x_5=0, x_3=1,	0
3	x_6	x_4	x_6=0, x_5=0, x_3=1,	0
4	x_1	x_5	x_6=0, x_1=0, x_3=1,	0
5	x_2	x_3	x_6=1, x_1=1, x_2=1/2,	-1/2
6	x_4	x_2	x_6=1, x_1=3/4, x_4=1,	-5/4

- Starting from iteration 4, the iterations are different when using Bland’s rule.

Anti-Cycling Rules – Lexicographic Method

- Dantzig et al. (1955) present Lexicographic Method which is to eliminate degeneracy, and finally prevent the cycling.
- Consider LP:

$$
\begin{aligned}
\text{minimize} \quad & c^T x \\
\text{subject to} \quad & a_1^T x = b_1 \\
& a_2^T x = b_2 \\
& \vdots \\
& a_m^T x = b_m \\
& x \geq 0
\end{aligned}
$$

- Assign a small positive constant ε_i to the RHS of i_{th} constraint with the order $0 << \varepsilon_m << \ldots << \varepsilon_2 << \varepsilon_1$.

Anti-Cycling Rules – Lexicographic Method

$$\begin{aligned}
\text{minimize} \quad & c^T x \\
\text{subject to} \quad & a_1^T x = b_1 + \varepsilon_1 \\
& a_2^T x = b_2 + \varepsilon_2 \\
& \quad \vdots \\
& a_m^T x = b_m + \varepsilon_m \\
& x \geq 0
\end{aligned}$$

- Adding such constants to the right hand side will ensure that there is a unique variable to leave the basis during an iteration
- The enter variable can be selected to be any non-basic variables with negative reduced cost.

Anti-Cycling Rules – Lexicographic Method

$$\begin{array}{ll} \text{minimize} & -x_1 - 2x_2 \\ \text{subject to} & 2x_1 + x_2 + x_3 = 30 \\ & x_1 + x_2 + x_4 = 20 \\ & x_1 + x_5 = 15 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{array}$$

$$\begin{array}{ll} \text{minimize} & -x_1 - 2x_2 \\ \text{subject to} & 2x_1 + x_2 + x_3 = 30 + \varepsilon_1 \\ & x_1 + x_2 + x_4 = 20 + \varepsilon_2 \\ & x_1 + x_5 = 15 + \varepsilon_3 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{array}$$

- Step 0: $$x_B^{(0)} = B^{-1}b = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \\ 15 \end{bmatrix} \qquad x_B^{(0)} = B^{-1}b = \begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 30+\varepsilon_1 \\ 20+\varepsilon_2 \\ 15+\varepsilon_3 \end{bmatrix}$$
- Step 1: $r_1 = r_2 = -1$, select x_1 as enter
- Step 2: $$d^1 = \begin{bmatrix} -B^{-1}N_1 \\ e_1 \end{bmatrix} = \begin{bmatrix} -\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

Anti-Cycling Rules – Lexicographic Method

- Step 3:

$$\alpha = \min\left\{\frac{30}{2}, \frac{20}{1}, \frac{15}{1}\right\} = 15 \qquad \alpha = \min\left\{\frac{30+\varepsilon_1}{2}, \frac{20+\varepsilon_2}{1}, \frac{15+\varepsilon_3}{1}\right\} = 15+\varepsilon_3$$

x_3 or x_5 can leave. Only x_5 can leave as $\varepsilon_3 << \varepsilon_1$.

Suppose x_5 leaves the basis.

- Step 4:

$$x_1 = \begin{bmatrix} 30 \\ 20 \\ 15 \\ 0 \\ 0 \end{bmatrix} + (15)\begin{bmatrix} -2 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 15 \\ 0 \end{bmatrix} \qquad x_1 = \begin{bmatrix} 30+\varepsilon_1 \\ 20+\varepsilon_2 \\ 15+\varepsilon_3 \\ 0 \\ 0 \end{bmatrix} + (15+\varepsilon_3)\begin{bmatrix} -2 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \varepsilon_1 - 2\varepsilon_3 \\ 5+\varepsilon_2-\varepsilon_3 \\ 0 \\ 15+\varepsilon_3 \\ 0 \end{bmatrix}$$

Degeneracy, may cycling. No degeneracy, no cycling

Anti-Cycling Rules – Lexicographic Method

- Theorem 3.3

  Assume that the constraint matrix of a linear program has full row rank m. If the leaving variable is determined by Lexicographic Method, then the Simplex Method will always terminate.

Revised Simplex Method

- Simplex Method require the basis B is invertible during iteration, this is computational expensive in practice.
- A better idea is to generate an inverse B^{-1} by solving an equivalent linear system of equations.
- For example, to compute the reduced costs the linear system $B^T\pi = c_B$ is solved first for and then the vector $r_N = c_N - \pi^T N$ can be easily computed.
- In practice, triangular factorization such as LU decom-position to solve the square system can enhance both the numerical stability and reduce space requirement

Revised Simplex Method

- Step 0: (Initialization)
  Generate an initial basic solution $x^{(0)} = [x_B \mid x_N]^{\mathrm{T}}$.
  Let B be the basis matrix and N be the non-basis matrix with corresponding partition $c = [c_B \mid c_N]^{\mathrm{T}}$.
  Let $\bar{B}$ and $\bar{N}$ be the index sets of x_B and x_N .
  Let $k = 0$, Go to Step 1.
- Step 1: (Optimality Check)
  Solve for π in the linear system $B^T\pi = c_B$.
  Compute the reduced cost $r_q = c_q - \pi^T N_q$ for all $q \in \bar{N}$.
  If $r_q \geq 0$ for all $q \in \bar{N}$, $x^{(k)}$ is optimal, STOP, else select one x_q from non-basis such that $r_q < 0$, Go to Step 2.

Revised Simplex Method

- Step 2: (Descent Direction Generation)

  Solve for d in the linear system $Bd = -N_q$.

  Construct $d^q = \begin{bmatrix} -B^{-1}N_q \\ e_q \end{bmatrix}$.

  If $d^q \geq 0$, then the LP is unbounded, else Go to Step 3.
- Step 3: (Step Length Generation)

  Compute the step length $\alpha = \min_{j \in \bar{B}} \left\{ -\frac{x_j^{current}}{d_j^q} \mid d_j^q < 0 \right\}$(the minimum

  ratio test). Let j^* be the index of basic variable

  then attains the minimum ratio α. Go to Step 4.
- Step 4: (Improved Adjacent Basic Feasible Solution)

  Let $x^{(k+1)} = x^{(k)} + \alpha d^q$. Go to Step 5.

Revised Simplex Method

- Step 5: (Basis Update)
  Let B_{j^*} be the column in B associated with the leaving basic variable x_{j^*}.
  Update the basis matrix B by removing B_{j^*} and adding the column N_q, thus

$$\bar{B} = \bar{B} - \{ j^* \} \cup \{q\}$$

  Update the non-basis matrix N by removing N_q and adding B_{j^*}, thus

$$\bar{N} = \bar{N} - \{q\} \cup \{ j^* \}$$

  Let k = k+1, Go to Step 1.

Example: Revised Simplex Method

$$
\begin{aligned}
\text{minimize} \quad & 2x_1 - x_2 \\
\text{subject to} \quad & -x_1 + x_2 + x_3 \phantom{{}+x_4} = 4 \\
& 2x_1 + x_2 \phantom{{}+x_3} + x_4 = 6 \\
& x_1 \ge 0, x_2 \ge 0, x_3 \ge 0, x_4 \ge 0
\end{aligned}
$$

Step 0: Initialization

$$
x^{(0)} = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 0 \\ 0 \end{bmatrix} \text{ with } B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ N = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}
$$

$$
c_B = \begin{bmatrix} c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},\ c_N = \begin{bmatrix} 2 \\ -1 \end{bmatrix},\ \bar{B} = \{3, 4\},\ \bar{N} = \{1, 2\}
$$

Example: Iteration 1

- Step 1: Solve $B^T\pi = c_B$ i.e.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \pi = \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$r_1 = c_1 - \pi^T N_1 = 2 - (0,0)\begin{bmatrix} -1 \\ 2 \end{bmatrix} = 2 \geq 0;\; r_2 = c_2 - \pi^T N_2 = -1 - (0,0)\begin{bmatrix} 1 \\ 1 \end{bmatrix} = -1 < 0$$

$x^{(0)}$ is not optimal, choose x_2 as enter. Go to Step 2

- Step 2: Solve $Bd = -N_q$ i.e.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = -\begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow d = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \qquad d^2 = \begin{bmatrix} d \\ e_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \not\geq 0$$

- Step 3: Compute step length

$$\alpha = \min_{j \in \bar{B} = \{3,4\}} \left\{ -\frac{x_j^{current}}{d_j^2} \mid d_j^2 < 0 \right\} = \left\{ -\frac{x_3^{current}}{d_3^2}, -\frac{x_3^{current}}{d_4^2} \right\} = \left\{ -\frac{4}{-1}, -\frac{6}{-1} \right\} = 4$$

. Go to Step 4.

Example: Iteration 1

- Step 4:

$$x^{(1)} = x^{(0)} + \alpha d^{(1)} = \begin{bmatrix} 4 \\ 6 \\ 0 \\ 0 \end{bmatrix} + (4)\begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ x_1 \\ x_2 \end{bmatrix}$$

x_3 leave the basis.

- Step 5: For $x^{(1)}$, $x_B = \begin{bmatrix} x_2 \\ x_4 \end{bmatrix}, x_N = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$

Update

$$B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, N = \begin{bmatrix} -1 & 1 \\ 2 & 0 \end{bmatrix}, c_B = \begin{bmatrix} c_2 \\ c_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, c_N = \begin{bmatrix} c_1 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\bar{B} = \{2, 4\}, \bar{N} = \{1, 3\}$$

Go to Step 1.

Example: Iteration 2

- Step 1: Solve $B^T\pi = c_B$ i.e.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow \pi = \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$r_1 = c_1 - \pi^T N_1 = 2 - (-1, 0)\begin{bmatrix} -1 \\ 2 \end{bmatrix} = 1 \geq 0; \;\; r_3 = c_3 - \pi^T N_3 = 0 - (-1, 0)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \geq 0$$

$r_N \geq 0$, STOP. $x^{(1)}$ is optimal.

Complexity of the Simplex Method

- Definition:

  A function $g(n) = O(f(n))$ (or Big-O of $f(n)$ if there is constant C > 0 such that for sufficiently large n)

  $$g(n) \leq Cf(n)$$

- E.g. the polynomial $g(n) = 2n^2 + 3n + 4 = O(n^2)$

  since $g(n) \leq 4f(n) = 4n^2$ for $n \geq 3$ where $f(n) = n^2$.

- For LP (or any problem), an algorithm is said to have polynomial time worst case complexity if the number of the operations required in the worst case is a polynomial function of the size of the problem.

- An algorithm is said to be exponential if the worst case complexity grows exponentially in the size of the problem.

Klee-Minty Problems

- Klee and Minty (1972) shows following LP with 2*m* variables and *m* constraints in standard form

$$\begin{aligned} &\text{minimize } \sum_{j=1}^{m} 10^{m-j} x_j \\ &\text{subject to } 2\sum_{j=1}^{i-1} 10^{i-j} x_j + x_i \le 100^{i-j}, \quad \forall i = 1,\dots,m \\ &\qquad\qquad x_1 \ge 0, \cdots, x_m \ge 0 \end{aligned}$$

- The simplex method has to enumerate all possible selection of *m* variables out of the 2*m* variables i.e. $\binom{2m}{m}$.
- If *m* = 50, then $\binom{100}{50} = 10^{29}$ that will take 3 trillion years to finished by a computer with capacity of 1 billion iterations per second.

Simplex Method MATLAB Code

- The standard LP can be solved by
  function [xsol objval iter exitflag]=SimplexMethod(c, Aeq, beq, B_set)
- Inputs:
  c = n$\times$1 vector, objective coefficient
  Aeq = m$\times$n matrix with m < n
  beq = m$\times$1 vector, RHS
  B_set = m$\times$1 vector, subscript set of basis
- Outputs:
  xsol = n*1 vector, final solution
  objval is a scalar, final objective value
  iter including all iteration details
  exitflag describes the exit condition of the problem as follows:
  0 - optimal solution
  1 - unbounded problem

Example 1: Bounded

- minimize $-x_1 - x_2$ minimize $-x_1 - x_2$

$$\begin{array}{llll} \text{subject to} & x_1 \le 1 & & \text{subject to } x_1 + x_3 = 1 \\ & x_2 \le 1 & \Rightarrow & x_2 + x_4 = 1 \\ & x_1 \ge 0,\ x_2 \ge 0 & & x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0,\ x_4 \ge 0 \end{array}$$

- Solve by following MATLAB code:

```
>> c = [-1; -1; 0; 0;];
>> Aeq = [1 0 1 0; 0 1 0 1];  Beq = [1; 1];
>> B_set = [3; 4]; % the subscript of basic variables
>> [xsol fval iter_detail exitflag]=SimplexMethod(c, Aeq, beq, B_set)
     probelm solved
      xsol = [1 1 0 0]^T
     fval = -2
     iter_detail = [1x1 struct] [1x1 struct] [1x1 struct]
     exitflag = 0
```

Example 2: Unboundedness

- minimize $-x_1 - x_2$
  subject to $-2x_1 + x_2 + x_3 \quad = 1$
  $$x_1 - x_2 \quad + x_4 = 1$$
  $$x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0,\ x_4 \geq 0$$
- Solve by following MATLAB code:

```
>> c=[-1; -1; 0; 0;];
>> Aeq=[-2 1 1 0; 1 -1 0 1];  beq=[1;1];
>> B_set=[3; 4]; % the subscript of basic variables
>> [xsol fval iter_detail exitflag]=SimplexMethod(c, Aeq, beq, B_set)
    unbounded problem
    xsol =  []
    fval =  []
    iter_detail = []
    exitflag = 1
```